29 — MKUFT Scientific Tightening and Claim Discipline

Professional reading edition of the evolving public MKUFT canon

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Scientific affiliation	Independent Researcher
Object status	Research module reading edition — not a standalone Zenodo publication
Public source	Open canonical public source
Source commit	`26e06b6fea52e6b052dbdd540e80d1d926b96b07`
MKUFT backbone DOI	10.5281/zenodo.17780566

Status boundary. This PDF is a reader-facing rendering of an evolving GitHub research module. It is not a frozen standalone paper and does not receive a separate DOI. The cited Git commit fixes the exact source identity of this edition.

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1. Purpose

MKUFT contains a broad architecture, mathematical scaffolds, experimental proposals, systems methods, and speculative branches. Breadth can create a false impression that every component carries the same evidential status.

This module separates four questions that must remain independent:

1. Is the architecture coherent?
2. Is the mathematics valid?
3. Is there empirical support?
4. Is there a physical mechanism?

Strength in one category does not silently transfer into another.

2. Architectural coherence

Architectural coherence concerns:

- typed S–I–P–O addressing;
- explicit relations and boundaries;
- reduction rules;
- falsifiers;
- promotion gates;
- controlled deformation;
- internal consistency.

A coherent architecture can be useful without being physically true.

3. Mathematical validity

Mathematical validity concerns:

- defined spaces and objects;
- dimensional consistency;
- normalisation;
- positivity where probabilities are used;
- valid operators;
- compatible units;
- declared approximation regimes;
- well-posed dynamics.

Valid notation does not by itself establish that nature uses the proposed model.

4. Empirical support

Empirical support concerns:

- preregistered predictions;
- adequate power and precision;
- strong ordinary nulls;
- blinding and instrumentation;
- held-out prediction;
- independent replication;
- effect-size stability;
- publication of null results.

A broad anomaly does not establish MKUFT unless the framework predicted a distinguishing fingerprint before the result was known.

5. Physical mechanism

A mechanism-level physical claim requires:

- physical state variables;
- units and observables;
- dynamical equations;
- couplings;
- symmetries and conservation conditions;
- recovery of accepted limits;
- at least one discriminating quantitative consequence.

A layer name, tuple, weighting term, projection, or named update map is not a completed physical mechanism.

6. Equation-status classes

E0 — notation or bookkeeping

E0 expressions keep concepts distinct without claiming a physical law. Examples include the S–I–P–O layer addresses, typed source and receiving spaces, or an index naming an experimental condition.

E1 — heuristic scaffold

An E1 expression is a mathematically legal candidate form that organises a hypothesis but has not been physically derived or empirically validated.

Representative E1 forms include an event-weighting scaffold,

$$\widetilde{W}(E) = \int D_{\text{phys}}(E \mid I)\, W(I \mid S, E)\, C(O \mid I, E)\, d\nu(I),$$

a provisional observer-condition perturbation,

$$C_O = C_0(1 + \varepsilon\kappa h),$$

a within-layer path cost,

$$C[\gamma] = \int_\gamma \lambda(x)\, ds,$$

and a generic boundary-response scaffold,

$$F_{\text{boundary}} = F_{\text{gradient}} + F_{\text{rotational}} + V_{\text{interaction}}.$$

These expressions may guide operationalisation. They are not established equations of nature merely because they can be written consistently.

E2 — operational statistical model

An E2 model has measurable variables, units or declared normalisation, frozen estimation rules, uncertainty treatment, and a defined dataset. It can be tested and compared with explicit baselines.

E3 — replicated empirical model

An E3 model is an E2 model that predicts held-out results and survives independent replication with stable parameter relationships. It supports the tested empirical relation only.

E4 — derived physical mechanism

An E4 model requires defined physical states, dynamics, couplings, symmetries, conservation conditions, accepted-limit recovery, and a discriminating quantitative prediction.

No current MKUFT foundational equation is classified as E4.

7. Novelty discipline

The use of familiar mathematics in new notation does not by itself constitute new physics.

Established ingredients used within MKUFT include measure and probability spaces, L^2 function spaces, graph and trajectory representations, Gibbs-like weighting, linear-response perturbations, anisotropic quadratic forms, threshold and bifurcation language, ablation, model comparison, and preregistration.

The strongest current novelty claim is architectural and methodological. Candidate contributions include:

- typed S–I–P–O layer-address discipline;
- explicit prevention of cross-layer evidence transfer;
- promotion gates from broad assay to discriminating signature;
- strongest-fair-null and complete-history replay controls;
- controlled deformation of supposedly load-bearing relations;
- integration of path cost, observer condition, recovery limits, and ontology boundaries within one research architecture;
- ATLD reciprocal-traversal testing as a functional systems hypothesis.

The appropriate current description is therefore:

> ***MKUFT proposes an original layered research architecture using established mathematical tools.***

That statement does not imply that the framework has already demonstrated a new fundamental field or physical law.

8. Claim ladder

Public claims can be located on the following ladder:

1. **Conceptual proposal** — an organised explanatory architecture.
2. **Mathematical scaffold** — candidate formal relations with explicit limitations.
3. **Operational hypothesis** — variables and measurements are defined.
4. **Discriminating prediction** — a locked fingerprint differs from strong alternatives.
5. **Replicated empirical relation** — independent confirmation exists.
6. **Mechanism-level result** — a parameterised mechanism explains and predicts the tested effect.
7. **Foundational physical theory** — accepted regimes are recovered and new quantitative physics is demonstrated.

Different MKUFT branches currently occupy different levels. A result in one branch does not promote the framework as a whole to a higher level.

9. Experimental priorities

Priority A — near-term operational flagship

The ATLD reciprocal-architecture programme tests active traversal against strongest-fair-null, complete-history replay, one-way, independent, reset, scrambled, and serious alternative-pairing controls. Its variables and controls can be operationalised without requiring speculative physics.

Priority B — foundational derivation programme

The Layer Before Law programme addresses the central physical claim. Bell compatibility, no-signalling, Born-rule recovery, QFT limits, gravitational recovery, and at least one new discriminator remain mandatory.

Priority C — controlled physical residual bridge

A productive physical test should use a defined apparatus, ordinary baseline, residual equation, parameter sweep, and predeclared response surface. Candidate domains include precision boundary conditions, dielectric, acoustic, photonic, fluid, plasma, or material-transition systems.

Priority D — reconnaissance assays

RNG/REG, remote-information, collective-correlation, and low-frequency physiological assays may remain exploratory branches or falsifiers. A generic positive anomaly would not by itself establish a unified field, a substrate, or observer-driven physical causation.

10. Observer-effect boundary

Ordinary quantum measurement does not require conscious observation as a physical cause. Measurement interaction and decoherence remain the baseline.

The MKUFT-specific observer question is narrower: whether a measured observer-state variable adds predictive value beyond apparatus, environment, expectation, and accepted physical models under preregistered controls.

11. Boundary and speculative-application firewall

The physics-facing core excludes unsupported aerospace, anomalous-craft, fixed-angle, and shape-specific motion claims from its evidential chain.

General boundary mechanics may examine measured gradients, anisotropy, threshold onset, stability, hysteresis, propagation, transition cost, and geometry-dependent response in controlled systems. Visual resemblance or remote reports do not establish a shared physical mechanism.

A speculative application becomes scientifically relevant only after the underlying mechanism is independently demonstrated and quantified in controlled systems.

12. Metaphysical and ethical-language boundary

Love, cohesion, dignity, grace, God, soul, and related concepts may remain philosophical, ethical, experiential, or systems-level elements of the wider framework. They are not converted into unmeasured physical variables and do not rescue failed physical predictions.

Where cohesion is quantified, it requires domain-specific operational definitions such as correlation, alignment, contradiction reduction, stability, network integration, error correction, or persistence under perturbation.

The ethical importance of a concept and its physical measurability are different questions.

13. Failure conditions

A physics-facing MKUFT branch is weakened or rejected where:

- additional variables cannot be operationally defined;
- its equations add no predictive value beyond accepted models;
- the relevant ordinary limit cannot be recovered;
- a claimed probability is non-normalisable or can become negative;
- dimensionful quantities enter exponentials without a declared scale;
- layer addresses cannot be distinguished operationally;
- an S-layer or O-layer term can be removed without changing any prediction attributed to it;
- a named mechanism remains only a placeholder after repeated attempts at formalisation;
- confirmatory predictions are selected after results are known;
- repeated adequately powered independent tests converge to the null;
- speculative examples are required to make the model appear predictive.

Failure of one branch does not become evidence for another branch and is not hidden inside the breadth of the framework.

14. Reader-facing scientific order

For a scientific claim, the clearest order is:

1. exact claim and current status;
2. nearest established frameworks;
3. specific proposed difference;
4. mathematical status;
5. operational variables;
6. strongest ordinary null;
7. locked prediction or derivation target;
8. failure condition;
9. what success would and would not establish;
10. wider philosophical interpretation after the evidence-bearing object is clear.

Breadth is best presented after the narrow claim and its evidence requirements are understood.

15. Current scientific position

MKUFT is presently best described as:

> ***an original, broad, pressure-testable layered research architecture with disciplined mathematical scaffolds, explicit failure conditions, and several operational research programmes, but without a completed or independently validated new fundamental physical mechanism.***

The route forward is discriminating derivation and experiment rather than stronger rhetoric.

Related public documents

- MKUFT Core Extended
- Mathematical Appendix
- Standalone Formal Addendum
- Experimental Test Programme
- Discriminating Experiments and Promotion Gates
- Falsification Summary
- Layer Before Law
- Scientific References and Current Literature